

Problem S5 – DNA Explorer 🚀

Difficulty

★ ★ Beginner

Overview

DNA is made up of four nitrogen bases:

- A (Adenine)
- T (Thymine)
- C (Cytosine)
- G (Guanine)

Scientists often examine different parts of a DNA sequence to study genes and mutations. Your task is to build a DNA Explorer that allows users to inspect different sections of a DNA sequence using Python slicing.

User Input

Use:

- `st.text_input()` → DNA Sequence

Example input:

ATCGGATCGTAGCTAGCTAGGCTAACCGTAGCTAGCTAGCTA

Validate the DNA Sequence

Ensure that the sequence contains only the following characters:

- A
- T
- C
- G

If any other character is entered, display an error message.

Buttons

Create the following buttons:

First 20 Bases Display the first 20 characters of the DNA sequence.

Last 20 Bases Display the last 20 characters.

Middle Section Display the middle 20 characters of the sequence.

Reverse Sequence Display the DNA sequence in reverse order.

Every Second Base Display every second character.

Every Third Base Display every third character.

DNA Length Display the total number of bases in the sequence.

Bonus Mission ★

Add two more analysis buttons:

Count Each DNA Base

Display:

A : 12

T : 11

C : 10

G : 9

DNA Preview

Display:

- First 10 bases
- Last 10 bases

This simulates how bioinformatics software previews long DNA sequences.